

Ameya Kulkarni

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EDUCATION

University of Washington, College of Engineering

Seattle, WA

Bachelor of Science: Mechanical Engineering

Expected June 2027

Cumulative GPA: 3.8/4.0

Dean's List x3, Theta Tau Professional Engineering Fraternity, NRG Research Group

Relevant Coursework (through Spring 2026): Mechanics of Materials • Kinematics and Dynamics • Differential Equations • Linear Algebra • Scientific Computing • Calculus • Physics • General Chemistry • Introduction to R • Java

RELEVANT SKILLS

Technical: CAD (SolidWorks, AutoCAD, Onshape, Revit), Ansys, MATLAB, Fluid x3D, Java, Python, Arduino, html

Other: Machine Learning (Random Forest, Logistic regression, Linear Regression, XG Boost), Django REST, Julia

WORK EXPERIENCE

Structural Integration Engineer

Seattle, WA

UW SARP (Society for Advanced Rocket Propulsion)

Sep 2025- Present

- Developed bulkheads and internal spars in **SolidWorks**, incorporating torsional and axial load analysis to assure structural integrity during liquid rocket firing. Fabricated aluminum alloy and composite parts in machine shop.
- Modeled a PLA avionics bay mount using **3D printing**, refining the shape to enhance accessibility and resistance.
- Programmed a **Python-based bearing failure calculator** integrating tensile, shear, and bolt shear models, reducing manual analysis time and streamlining structural analysis workflows; presented methodology at **CDR**.

Research Assistant

Seattle, WA

Novosselov Research Group (NRG)

Oct 2025-Present

- Assisted graduate researchers with a study on electrohydrodynamic flow control, analyzing how electrode configurations influence boundary layer behavior, airflow stability, and thrust generation in low-speed regimes.
- Operated **wind tunnels** and utilized pitot tubes and pressure sensors to collect, calibrate, and process velocity and pressure data across various electrode configurations using **Python** and **MATLAB** for post- processing.
- Processed and analyzed test data to evaluate airflow and thrust changes caused by applied electric fields.
- Collaborated with graduate researchers to design modular test fixtures and refine experimental procedures to improve repeatability, measurement accuracy, and data consistency across multiple trials.

Structural Engineer

Seattle, WA

Husky Satellite Lab: University of Washington

Nov 2024 - Mar 2025

- Designed and optimized CubeSat frame components including custom hinge mechanisms using **SolidWorks**, conducting vibration and stress simulations (**static FEA**) on **Ansys** to improve structural stability by **4.5%**.
- Used Machine Shop Certification to manufacture precise components with tight tolerances utilizing manual **mills, lathes**, and 3 axis CNC machinery while simplifying production by establishing manufacturing schedule.
- Applied **CAD** to create and **Ansys Thermal** to simulate heat flow through core spacecraft subsystems, including battery packs and PCB assemblies, to evaluate and engineer clear temperature regulation strategies.
- Created documentation standards for the Structures team on Notion, including simulation reports, CAD modification protocols, and version control guidelines to ensure consistency and traceability for future teams.

Aircraft Systems Laboratory Researcher

Irvine, CA

UC Irvine x GATI: BEAM Program

Jun-Aug 2024

- Researched the influence of rotor geometry on aeroacoustics to suggest propeller modifications for aircraft.
- Designed various airfoils of propeller shapes in **SolidWorks** and ran trials in **Fluid x3D** to visualize drag.
- Developed an experiment using **Arduino** testing 2, 3, and 4 bladed propellers measuring **sound pressure level** and **tonal noise** while toggling with motor potentiometer for accurate measurements in **wind tunnel**.
- Presented findings to UCI grad students and Dr. Jacqueline Huynh to draft implications for the FAA.

ADDITIONAL INFORMATION

- **Hiller Aviation Museum:** Manned drone, flight simulators, guided tours, exhibition maintenance.
- **Part Time Instructor at Communication Academy:** Speech therapy for students ages ranging from 9-12.
- **UW Badminton Team:** Competitive team finished 6th place in the 2025 National Collegiate Championships.